PRACTICAL 8

Building a RESTful API using Node.js and Express.js

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Program / Semester	3RD
Course / Subject	WEB DEVELOPMENT FRAMEWORK
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Submission Date	28-10-2026

1. Aim

To build a server-side web application and create RESTful API endpoints (GET, POST, PUT, DELETE) using Node.js and the Express.js framework.

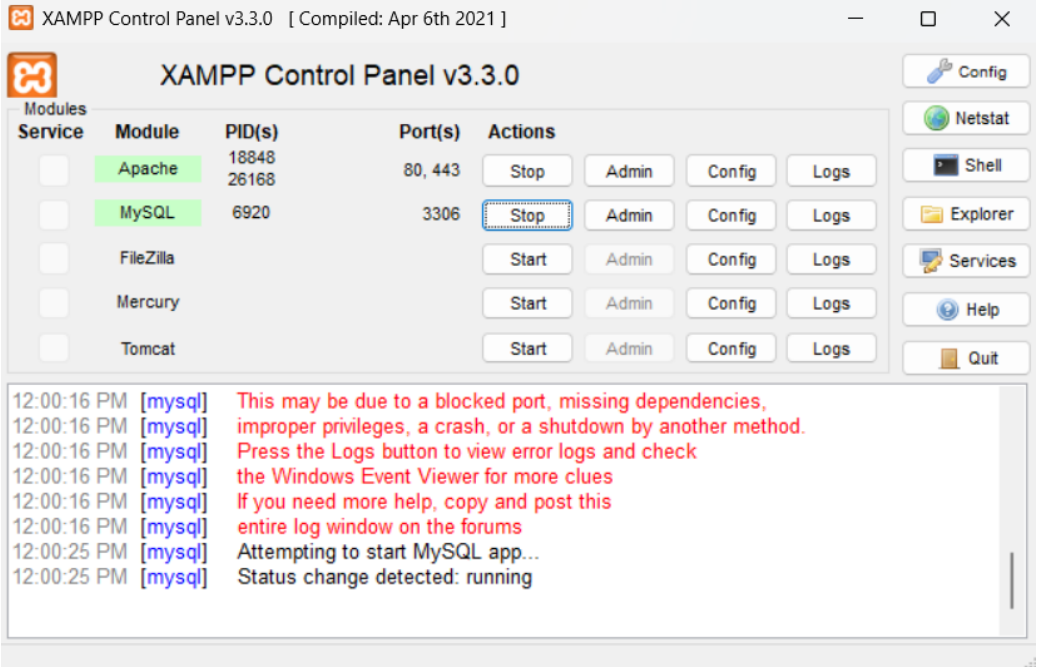
2. Practical Requirements

- schema.sql: SQL dump with 3NF tables (students, events, registrations), PK/FK keys, and test data.
- db.php: Secure PHP PDO connection file with try-catch error handling.
- test_connection.php: PHP code fetching data using Prepared Statements.

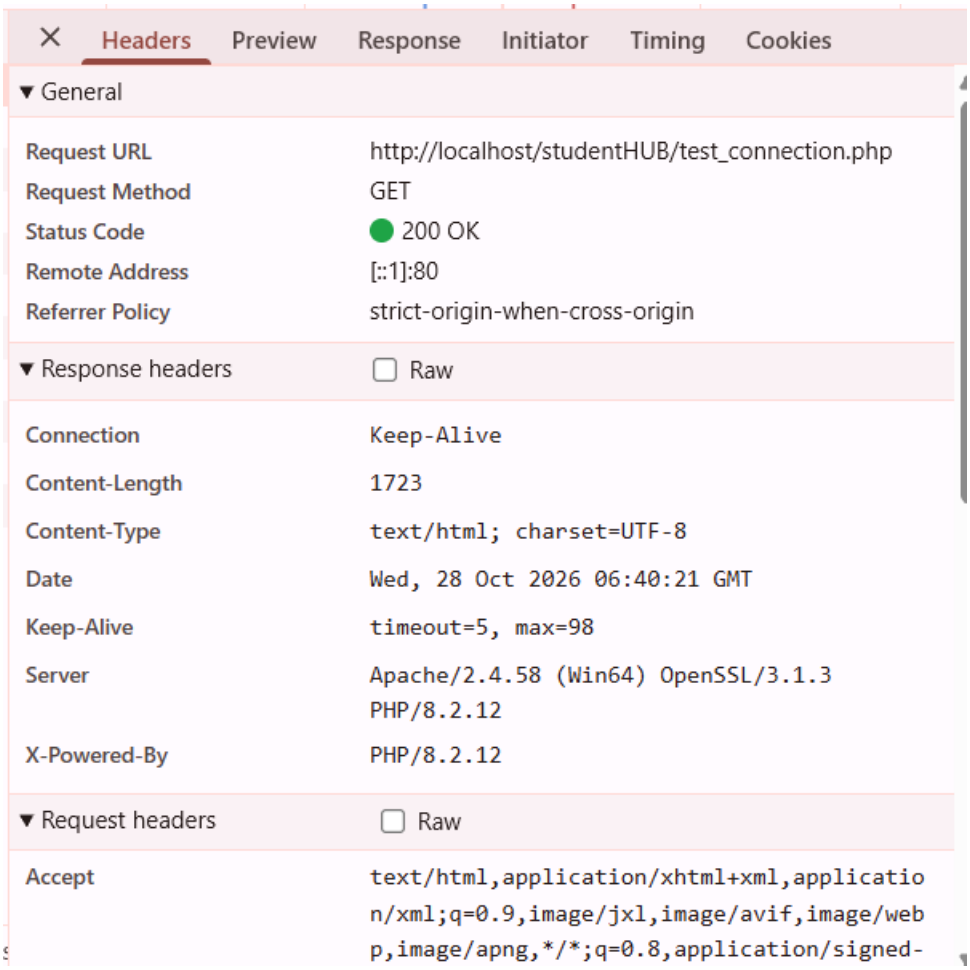
3. Required Screenshots / Demonstration

Insert the screenshots below in the same order as the practical instructions.

3.1 Server Initialization & Running



3.2 GET Request Output



3.3 Table Showing

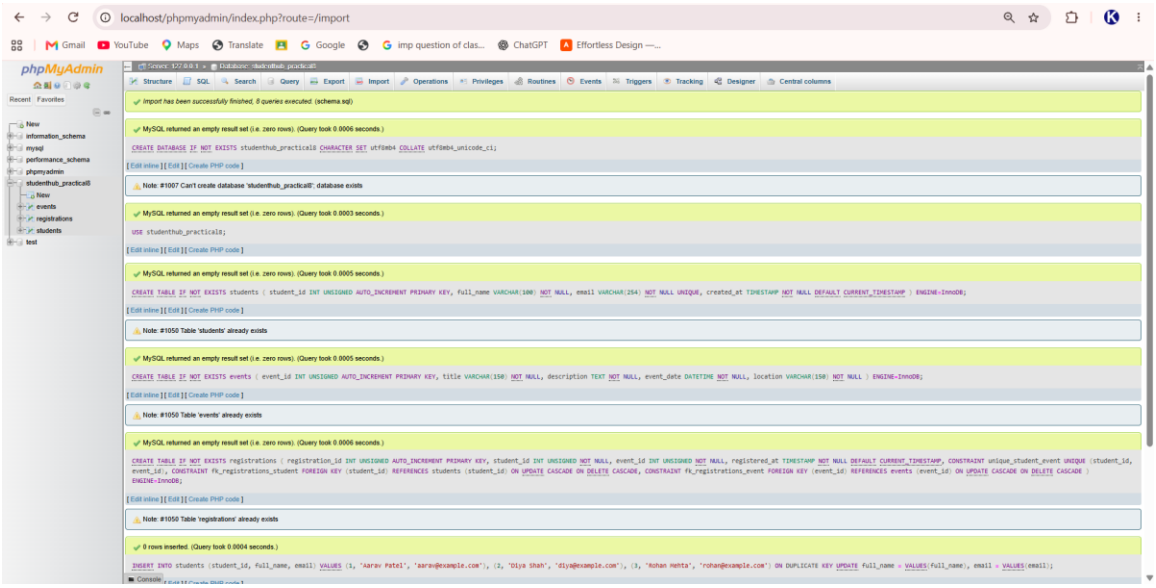
Practical 8: Database Connection

Database connection successful. Registration data is displayed below.

Web Development Workshop Registrations

Registration ID	Student	Email	Event	Registered At
1	Aarav Patel	aarav@example.com	Web Development Workshop	2026-10-28 12:01:55
2	Diya Shah	diya@example.com	Web Development Workshop	2026-10-28 12:01:55

3.3 SQL Query Import (Database Already Exist)



4. Result

The RESTful API was successfully created using Node.js and Express.js. All CRUD operations (Create, Read, Update, Delete) were configured with appropriate routing. The endpoints were tested and verified to handle client requests, process JSON payloads, and return the correct HTTP status codes and responses.

5. Conclusion

This practical demonstrates the core concepts of server-side web development using the Express.js framework. By building RESTful endpoints, the foundational backend architecture required for full-stack web applications was established, improving overall understanding of client-server data exchange and modular backend routing.

— End of Practical 8 Submission —